



ME 300:
Engineering Thermodynamics

Example: Heat
Exchangers

Example: Heat exchanger

Water is heated by air in a heat exchanger. The water enters at 150 C and 0.2 MPa and leaves at 300 C and 0.2 MPa. The mass flow rate of the water is 1 kg/s. The air enters at 350 C and 0.1 MPa at a rate of 2 kg/s. Determine the heat-transfer rate between the two fluids and the exit temperature of the air.

H₂O

$$T_1 = 150^\circ\text{C}$$

$$P_1 = 0.2 \text{ MPa}$$

$$T_2 = 300^\circ\text{C}$$

$$P_2 = 0.2 \text{ MPa}$$

$$\dot{m}_{\text{H}_2\text{O}} = 1 \text{ kg/s}$$

Air

$$T_1 = 350^\circ\text{C}$$

$$P_1 = 0.1 \text{ MPa}$$

$$\dot{m}_{\text{air}} = 2 \text{ kg/s}$$

Find

$$\dot{Q}$$

$$T_{2,\text{air}}$$

Assume

- steady-flow

- H₂O → S.C.S.

- air → ideal gas

- $\Delta ke = \Delta pe = 0$, $\dot{W} = 0$

- quasi-equilibrium.

$$\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \dot{Q} - \cancel{\dot{W}}$$

$$\dot{m}\Delta h = \dot{Q}$$



Which of the following assumptions is *NOT* appropriate for the water?

A. Quasi-equilibrium

B. $\Delta KE = \Delta PE = 0$

☒ C. Ideal gas

D. They are all appropriate

Example: Heat exchanger

Water : $P_1 = 0.2 \text{ MPa}$
 $T_1 = 150^\circ\text{C} = 423 \text{ K}$ } NIST

D2 $\rightarrow T_{\text{sat}} @ 0.2 \text{ MPa} = 393 \text{ K} < T_1$
 $\Rightarrow h_1 = 2768.8 \frac{\text{kJ}}{\text{kg}}$

State 2 : $P_1 = 0.2 \text{ MPa}$

$T_2 = 300^\circ\text{C} = 573 \text{ K}$

$\Rightarrow h_2 = 3071.8 \frac{\text{kJ}}{\text{kg}}$

$$\dot{m} \Delta h = \dot{Q}$$

$$\dot{Q} = \dot{m} (h_2 - h_1) \quad \sqrt{\frac{\text{kJ}}{\text{kg}}}$$
$$= (1) (3071.8 - 2768.8)$$

$$\dot{Q} = 303 \text{ kW}$$

Example: Heat exchanger

$$\dot{Q}_{H_2O} = -\dot{Q}_{air} \Rightarrow \dot{Q}_{air} = -303 \text{ kW}$$

$$\Rightarrow \dot{m}(h_2 - h_1) = \dot{Q}_{air} \quad \rightarrow \text{ideal, const. } c_p$$

$$\dot{m}c_p(T_2 - T_1) = \dot{Q}_{air}$$

$$T_2 = T_1 + \frac{\dot{Q}_{air}}{\dot{m}c_p} = 623 + \frac{(-303000)}{(2)(1001)} \Rightarrow \boxed{T_2 = 477.4 \text{ K}}$$